Homework Assignment 3:
Compressible Flow

Mech 588 — Spring, 2023

Due Date: April 3

There are several places in the notes where I cheerfully assert that something is true without any proof. This question will close some of those gaps.

1. (3 marks) For the analytic flux for 1D compressible, inviscid flow, show that

$$F = \frac{\partial F}{\partial U} U$$

2. (5 marks) For Roe's scheme, with left and right states U_L and U_R, respectively, show that

$$F_L - F_R = \tilde{A} \cdot (U_L - U_R)$$

3. (7 marks) Suppose you were writing a program to solve compressible, inviscid flow problems and you wanted to use implicit time advance. Not only would you need to be able to calculate the flux Jacobians, but you would need to assemble them into a block tridiagonal matrix.
   For a one-dimensional, uniform mesh with spacing Δx, write out the fully-discrete system you would use for Roe's flux-difference splitting. Be sure to use second-order spatial accuracy for the flux integral. Use the implicit Euler time advance scheme. When you compute the left-hand side matrix, use only first-order spatial accuracy (in other words, compute the Jacobians, etc, for the first-order accurate spatial discretization. You may neglect variations in the analytic flux Jacobian A (and its parts). Feel free to simply write A or $\tilde{A}$ for the Jacobians, and to define expressions like $U_{i+\frac{1}{2};i}$ to abbreviate the extrapolated solution at the interface.